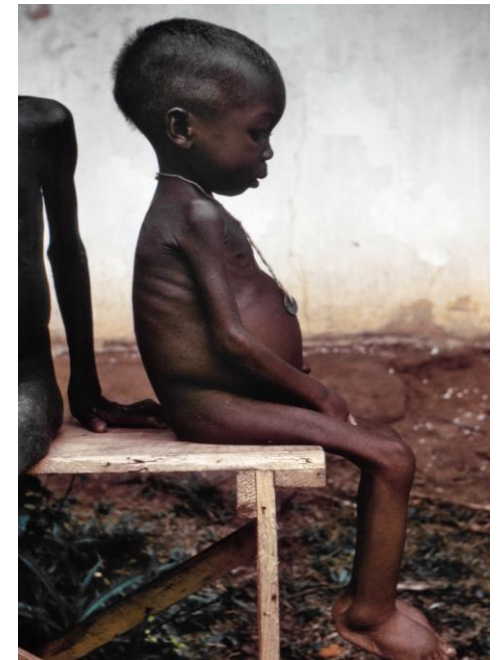


Protein deficiency diseases

Kwashiorkor stems from severe protein deficiency despite enough calories, often from diets high in carbohydrates (rice, maize) and low in protein-rich foods.

- **Causes (Sources)**
- **Insufficient Protein:** Diets lacking protein, common in areas relying heavily on starchy staples like cassava, rice, or maize.
- **Weaning Practices:** Often seen in children weaned onto low-protein diets when a new sibling arrives.
- **Infection:** Recent infections (like measles) can trigger or worsen it.
- **Socioeconomic Factors:** Poverty, food insecurity, and cultural practices.



Protein deficiency diseases

- **Symptoms (Signs)**

1. **Edema:** Swelling, especially in legs, feet, and face (a bloated belly).
2. **Skin & Hair:** Dry, peeling skin, rashes, brittle, discolored hair.
3. **Growth Issues:** Stunted growth, poor weight gain, loss of muscle mass.
4. **Behavioral Changes:** Irritability, lethargy, apathy, fatigue.
5. **Weak Immunity:** Frequent infections.

- **Treatment: Nutritional Therapy of Kwashiorkar**

- **Gradual Refeeding:** Slowly increase calories and protein, often with milk-based formulas or ready-to-use therapeutic foods (RUTF).
- **Supplements:** Essential vitamins and minerals.
- **Balanced Diet:** Gradually introduce protein-rich foods (eggs, meat, legumes).

Protein deficiency diseases

Marasmus stems from severe lack of calories/protein, causing extreme weight loss, muscle wasting, sunken eyes, and dry skin; symptoms include weakness, irritability, and stunted growth.

Source (Causes)

- **Inadequate Intake:** Primarily from a diet lacking enough calories and protein, often due to poverty, famine, drought, or poor access to nutritious food.
- **Early Cessation of Breastfeeding:** Stopping breastfeeding too early and replacing it with diluted formula. Underlying Conditions: Chronic illnesses that hinder nutrient absorption.



Protein deficiency diseases

Symptoms

Physical: Severe weight loss, emaciated appearance, absence of fat/muscle, sunken eyes, wrinkled skin, dry/thin hair, prominent bones, stunted growth.

Behavioral: Weak cry, irritability, lethargy, reduced activity.

Other: Weakened immune system (frequent infections), dehydration, electrolyte imbalance, delayed development.

Treatment (Phased Approach)

Stabilization (Initial Phase): Correct dehydration (ReSoMal), electrolyte issues, and hypothermia; treat infections (antibiotics); focus on preventing refeeding syndrome.

Rehabilitation (Gradual Refeeding): Slowly increase calorie/protein intake with nutrient-rich, milk-based formulas, potentially via tubes.

Follow-Up: Long-term dietary education for the family, monitoring, and ensuring continued balanced nutrition.

Vitamin deficiency overview

Vitamins	Deficiency disorder
Vitamin A (Retinol)	Loss of vision (Blindness).
Vitamin B ₁ (Thiamine)	Weak muscles and severe weight loss (Beriberi).
Vitamin B ₂ (Riboflavin)	Poor diet and mouth sores (Ariboflavinosis).
Vitamin B ₃ (Niacin)	Diarrhea and dermatitis (Pellagra).
Vitamin B ₅ (Pantothenic acid)	Ischemic attacks and multiple sclerosis affect brain (Paresthesia).
Vitamin B ₆ (Pyridoxine)	Low number or lack of red blood cells (Anemia).
Vitamin B ₇ (Biotin)	Red and itchy skin (Dermatitis).
Vitamin B ₉ (Folic acid)	Immature RBCs in the bone marrow (Megaloblastic anemia).
Vitamin B ₁₂ (Cyanocobalamin)	Abnormal absorption in the small intestine (Pernicious anemia).
Vitamin C (Ascorbic Acid)	Bleeding gums and swollen joints (Scurvy).
Vitamin D (Calciferol)	Weakening of bones near joints (Rickets).
Vitamin E (Tocopherols)	Faster depletion of RBCs from the body (Hemolytic anemia).
Vitamin K (Phylloquinone)	Irregular and slow blood bleeding and clotting (Bleeding diathesis).